Def'n of a cont. fnc. using open sets.

Proposition: Let $f: X \to Y$ be a function. Then the following are equivalent:

(i) f is continuous (i.e., a map)

(ii) Let $f(x) = y$. Then, given an open $y \in \mathcal{O} \subset Y$ there is an open $x \in U \subset X$ such that $f(U) \subset \mathcal{O}$. local continuity

Pf: (i) $\to$ (ii). Let $y \in \mathcal{O} \subset Y$ be open. Then $x \in f^{-1}(\mathcal{O}) \subset X$ is open and $f(f^{-1}(\mathcal{O})) \subset \mathcal{O}$. Thus (ii) holds with $U = f^{-1}(\mathcal{O})$. HW 5, 15

(ii) $\to$ (i). Let $\mathcal{O} \subset Y$ be open. If $f(X) \cap \mathcal{O} = \phi$, then $f^{-1}(\mathcal{O}) = \phi$, which is open by definition. If $f(X) \cap \mathcal{O} \neq \phi$, pick any $x \in f^{-1}(\mathcal{O})$. Then, by (ii), there is $x \in U_x \subset X$ open such that $f(U_x) \subset \mathcal{O}$. Do this for all $x \in f^{-1}(\mathcal{O})$. It follows that $f^{-1}(\mathcal{O}) = \bigcup_{x \in f^{-1}(\mathcal{O})} U_x$, which is open. Thus (i) holds. so $U_x \subset f^{-1}(\mathcal{O})$

Notation: $y \in \mathcal{O} \subset Y$ (or $\underset{y}{\mathcal{O}} \subset Y$) reads: "$\mathcal{O}$ is open in Y, y is contained in $\mathcal{O}$." That is, $\mathcal{O}$ is an open nbhd of y.

Ex: If $f: \mathbb{R} \to \mathbb{R}$, then what is condition (ii) known as?

Let $f: X \to Y$, $g: Y \to Z$ be continuous.
Then $g \circ f: X \to Z$ is continuous.

Proof on p. 8.

A map $f: X \to Y$ is

(i) open if $f(O) \subset Y$ is open for all $O \subset X$ open

(ii) closed if $f(C) \subset Y$ is closed for all $C \subset X$ closed

(iii) a homeomorphism if there is some $g: Y \to X$ such that $g \circ f = id_X$, $f \circ g = id_Y$.

Note: A bijection $f: X \to Y$ is a homeom. if and only if it is open, or, if and only if it is closed.

Pf: Let $\mathcal{O} \subset Z$ be open. Then $g^{-1}(\mathcal{O}) \subset Y$ is open and so is $f^{-1}(g^{-1}(\mathcal{O}))$. But $(g \circ f)^{-1}(\mathcal{O}) = f^{-1}(g^{-1}(\mathcal{O}))$.

Thm: Let $f: X \to Y$ be cont, and let $A \subset X$ be equipped with the subspace topology. Then the restriction $f|_A : A \to Y$ of f to A is continuous.

Pf: Let $\mathcal{O} \subset Y$ be open. Then

$$(f|_A)^{-1}(\mathcal{O}) = A \cap f^{-1}(\mathcal{O})$$

is open in A.

Thm: The following are equivalent:

(a) $f: X \to Y$ is continous

(b) If $\mathcal{B}$ is a basis for the topology of Y, then $f^{-1}(B) \subset X$ is open for all $B \in \mathcal{B}$.

(c) $f(\bar{A}) \subset \overline{f(A)}$ for any $A \subset X$

(d) $\overline{f^{-1}(B)} \subset f^{-1}(\bar{B})$ for any $B \subset Y$

(e) $f^{-1}(C) \subset X$ is closed for any closed $C \subset Y$.

HW3 2.c 1.d

(a) $\leftrightarrow$ (b) easy. ($f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$ etc.)

(a) $\leftrightarrow$ (e): $f^{-1}(Y \setminus B) = X \setminus f^{-1}(B)$

(e) $\leftrightarrow$ (d): Assume (e). Then $f^{-1}(\bar{B})$ is closed and contains $f^{-1}(B) \Rightarrow \overline{f^{-1}(B)} \subset f^{-1}(\bar{B})$

Next assume (d), and let $B \subset Y$ be closed. Then by (d), $f^{-1}(\bar{B}) = f^{-1}(B) \supset \overline{f^{-1}(B)}$. It follows that $f^{-1}(B)$ is closed.

(e) $\leftrightarrow$ (c) Assume (e): Then $f^{-1}(\overline{f(A)})$ is closed and contains A so $\bar{A} \subset f^{-1}(\overline{f(A)})$. Hence $f(\bar{A}) \subset \overline{f(A)}$

Assume (c): Let $B \subset Y$ be closed. Then by (c), $f(\overline{f^{-1}(B)}) \subset \overline{f(f^{-1}(B))} \subset \bar{B}$. But $\bar{B} = B$ by assumption. Hence $\overline{f^{-1}(B)} \subset f^{-1}(B)$, so $f^{-1}(B)$ must be closed.

Example: Let $S^n \subset \mathbb{R}^{n+1}$ be the unit sphere.
Define the stereographic projection $S^n \setminus \{p\} \to \mathbb{R}^n$,
where $p = (0, 0, \dots, 0, 1)$ is the north pole by

$$(y_1, y_2, \dots, y_n) = \left(\frac{x_1}{1 - x_{n+1}}, \frac{x_2}{1 - x_{n+1}}, \dots, \frac{x_n}{1 - x_{n+1}} \right) = \varphi(x_1, \dots, x_{n+1})$$

This is continuous as $\varphi : \mathbb{R}^{n+1} \setminus \{p\} \to \mathbb{R}^n$ is continuous.

Inverse: $r^2 = y_1^2 + y_2^2 + \dots + y_n^2 = (x_1^2 + x_2^2 + \dots + x_n^2) / (1 - x_{n+1})^2$

$$= (1 - x_{n+1}^2) / (1 - x_{n+1})^2 = \frac{1 + x_{n+1}}{1 - x_{n+1}}$$

$$\Leftrightarrow \quad x_{n+1} = \frac{-1 + r^2}{1 + r^2}; \quad 1 - x_{n+1} = \frac{2}{1 + r^2}$$

Hence $x_i = 2y_i/(1+r^2)$, $1 \le i \le n$.

So $\varphi^{-1}: \mathbb{R}^n \to S^n \setminus \{p\}$ is continuous.

Prop: Let $f: X \to Y$ and suppose $f(X) \subset A \subset Y$. Equip A with subspace topology. Then $f: X \to A$ is cont.

Pf: Let $B \subset A$ be open. Then $B = A \cap O$, $O \subset Y$ open. Now $f^{-1}(B) = f^{-1}(A \cap O)$
$= f^{-1}(A) \cap f^{-1}(O) = X \cap f^{-1}(O) = f^{-1}(O)$, which is open in X.

Def'n: Let (X, τ) be a topological space. Call X Hausdorff if given $x \neq y$, then there are nbhds U_x, U_y of x and y s/t $U_x \cap U_y = \phi$.

Ex: Any metric space.

Ex: Real line with a point added is not H.

Ex: Order topology is H.

Ex: Lower limit topology on $\mathbb{R}$

Ex: Sierpinski space: $X = \{0,1\}$, $\tau = \{\phi, \{1\}, \{0,1\}\}$

Note: H-spaces are also called T_2.

Pts are closed

Theorem: The following are equivalent:

1. X is H.
2. Given $x \neq y \in X$, there is a nbhd U of x s/t $y \notin \overline{U}$.
3. Given $x \in X$, let $\mathcal{N}_x$ denote the collection of all nbhds of x. Then $\bigcap_{U \in \mathcal{N}_x} \overline{U} = \{x\}$.

Pf: $1 \to 2$. Given $x \neq y$, there are disjoint nbhds $x \in U_x$, $y \in U_y$. Then $U_x \subset \neg U_y$, which is closed, so $\overline{U_x} \subset \neg U_y \Rightarrow y \notin \overline{U_x}$.

$2 \to 3$: If $y \neq x$, then $y \notin \overline{U}$ for some nbhd U of x. Hence $y \notin \bigcap_{U \in \mathcal{N}_x} \overline{U}$.

$3 \to 1$ Given $y \neq x$, then $y \notin \bigcap_{U \in \mathcal{U}_x} \overline{U} \Rightarrow y \notin \overline{U}$ for some $U \in \mathcal{U}_x$. Then U, $\overline{U}$ are disjoint nbhds of x, y.

Prop: Let $A \subset X$, X be H. Then A is H.

Prop: Let X be H. Then any finite set is closed. (Only need to show that every singleton $\{x\} \subset X$ is closed.)

Prop: Let $A \subset X$, where X is H. Then $x \in A'$ iff $U \cap A$ is infinite for any nbhd U of x.

Pf: Obviously, the condition is sufficient. Now suppose that $U \setminus \{x\} \cap A = \{a_1, \ldots, a_k\}$ for some nbhd U of x. Now $V = U \cap X \setminus \{a_1, \ldots, a_k\}$ is a nbhd of x and $V \setminus \{x\} \cap A = \phi$. Thus $x \notin A'$.

Def'n: Suppose X is H. (or, alternatively, every $\{x\} \subset X$ is closed). Call X normal (or T_4) if given disjoint closed A, B, then there are disjoint open sets containing A and B, respectively.